implementation of a queue, removal in the middle of the queue takes linear time).

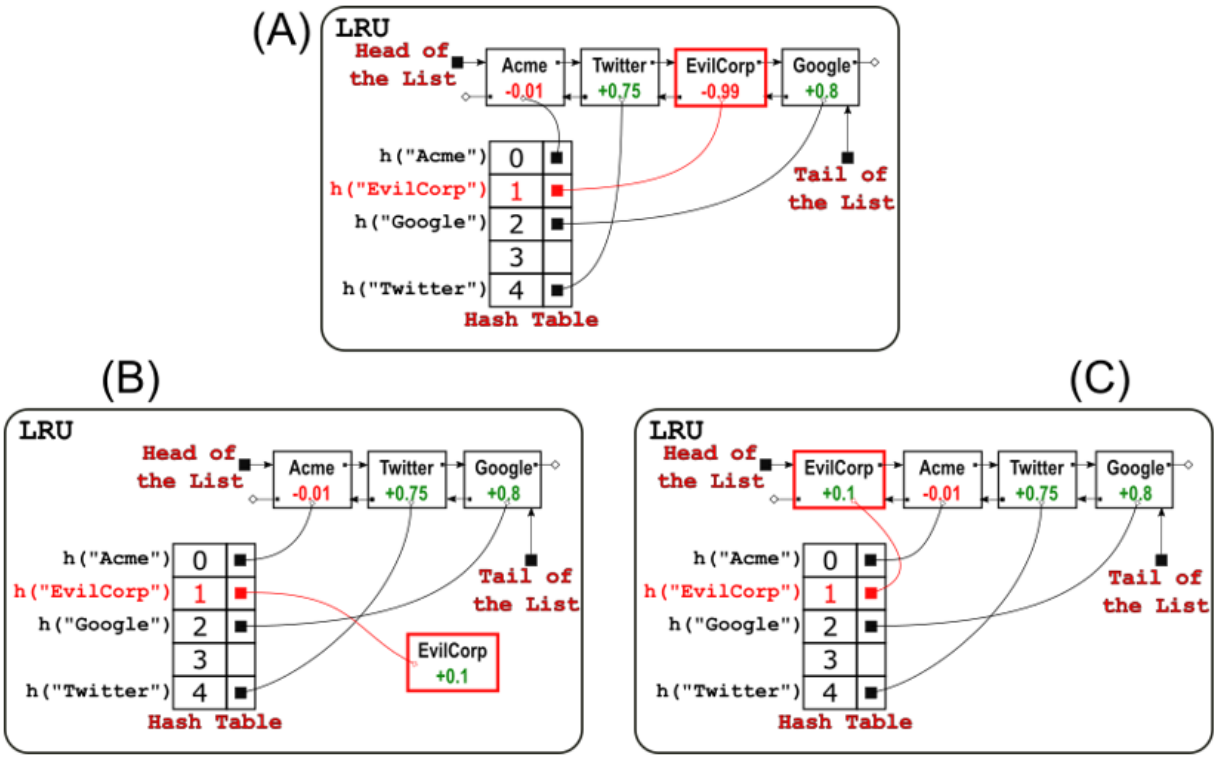


Figure 7.7 Update of a cache entry on cache hit.
(A) The initial state of the cache. At this stage, the entry to update, “EvilCorp”, has been looked up and we had a cache hit. The lookup happens on the hash table hence its entry and link are highlighted in red.
(B) The list node for EvilCorp is updated with the new data (looks like EvilCorp spent some *cash* to turn around its reputation!) and its also removed from the list, whose links are updated.
(C) The node is now added to the front of the list. No update is necessary for the hash table.

If the cache is full, we need to remove the least recently used entry before we can add a new one. In this case, we